Examiner only

6. *(a)* State the conditions necessary for a body to remain in equilibrium. [2]

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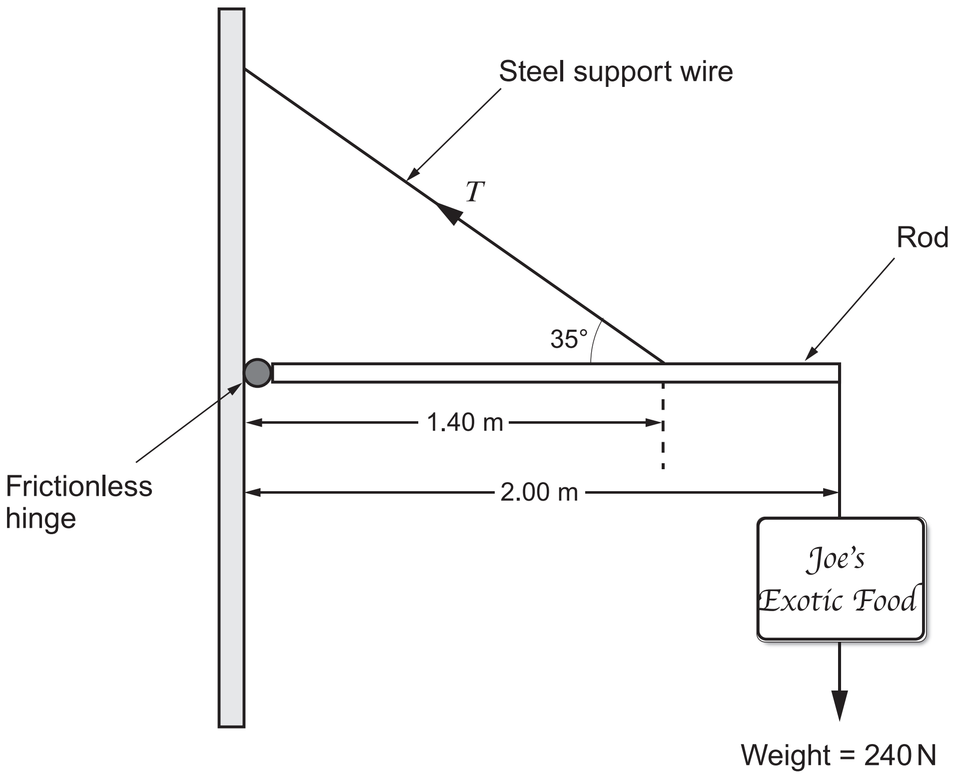
(b) A shopkeeper wishes to hang a heavy sign of weight 240 N outside his shop using the following arrangement. He has steel wires of various diameters to choose from. You may neglect the weight of the rod.



The shopkeeper needs to ensure that the steel wire he decides to use has a breaking strength greater than is required to support the sign. By taking moments about an appropriate point, show that the tension, T, in the wire in the above diagram is approximately 600 N. [3]

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Examiner only

(c) The shopkeeper finds the following information on a website advertising steel wire ropes.

Minimum breaking strength and safe working load of uncoated steel wire ropes are indicated below.

Diameter of rope / mm	Minimum breaking strength / N	Safe working load / N at 5:1 ratio	Safe working load / N at 3:1 ratio
1.5	1 900	380	633
2.0	2 750	550	916
2.5	3 300	660	1 100
3.0	5 400	1 080	1 800

Loading Information

SAFE WORKING LOAD (SWL) is the load that can be applied without causing any damage to the wire rope.

WE PROVIDE TWO SWLs FOR YOUR CONSIDERATION AT RATIOS 5:1 AND 3:1.

Factors of safety should always be applied when determining maximum wire rope loading conditions. If in doubt a suitably qualified engineer should be consulted to assess loading factors.

Examiner only

(i) Based on the information on the web page the shopkeeper decides to apply a SWL ratio of 3:1 for the wire he will use. State, giving your reasoning, which **minimum** diameter of wire rope he should choose to use. [2]

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(ii) The shopkeeper has no engineering background. Evaluate whether or not he has made an informed decision. [2]

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TURN OVER FOR THE LAST QUESTION